

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 202 — Past Questions & Materials
by Renance

Course: MTH 202
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

This study pack was compiled and restructured by Renance for MTH 202 students at Federal University of Petroleum Resources, Effurun. It brings together 3 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 3 — MTH 202/221 ordinary differential equations 2025

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MATHEMATICS _ ee

B.Sc. (MATHEMATICS) DEGREE EXAMINATION (2024/2025), FIRST SEMESTER.

MTH 202/224 (ORDINARY DIFFERENTIAL EQUATIONS 1 3C). September, 2025. oo

Instructions: Answer any four (4) questions. ie

Credit shall be given for brevity and concise answers.

Consider the solution of the ordinary differential equation (ODE);

$$y^2 C_1 e^{5x} + C_2 e^{3x},$$

i. Obtain the ordinary differential equation (ODE) from its solution above.

(10marks)

ii. State the degree and order of the ordinary differential equation (ODE) obtained?

(5marks)

iii. In the process of obtaining the differential equation from its solution, what is the essence of differentiating with respect to x? or a (2.5marks)

Question (2 ;

Given the Bernoulli's equation;

$$\frac{dy}{dx} + \frac{y}{x} = 5y^2$$

ax

i. Reduce the equation to a first-order linear differential equation. (6marks)

(6marks)

Hence, obtain the solution of the equation.

Briefly explain the terms "Order" and "Degree" of differential equation with examples to back up your explanation. (5.5marks)

Question (3)

Solve for the solution of the differential equation (7marks)

eg d

$$(6x^5 + 4x^3y^3 - 3xy^2t) = (3x^4y^2 - 4x^3y^3 - 5y^5)$$

il.

ili.

il. Solve the ordinary differential-equation (7marks)

$$\frac{dy}{dx} - 2xy$$

$$dx = 1+x^2 \quad \text{---}$$

iii. | Differentiate between ordinary differential equation (ODE) and partial differential equation (PDE) (3.55marks)

QUESTION 4

Find the general solution of the following differential equations

| eae o#y _ 332% 4121y =0 (5.5 marks)

dx² dx

ii. @y . 96% 4 169y =9 (5.5 marks)

ax? ax

(6.5 marks)

iii. ay 44 =0

cat ty, t oF

QUESTION 5

(5 marks)

i. Show that $y_1 = e^*$ and $y_2 = e^%$ are linearly independent

ii. Prove that $L_7^*f(0) = sf(t) - f''@$ (7.5 marks)

aes _ \$417

iii. Evaluate £7* (=) (5 marks)

QUESTION 6

i. Solve the IVP $2y'' - 15y' + 25y = 9$, $y(0) = 1, y'(0) = 2$ using Laplace transform method (10.5 marks)

(3.5 marks)

ition principle

function $f(t)$ and hence find £L(1 — e737")

(3.5 marks)

State without proof, the Superpos

ii.

Define the Laplace transform of a

Paper 2 of 3 — MTH 202 Elementary Differential Equations 2020 2021 Second Semester

Level 200 • Second Semester • PastQuestion

Instructions:

1. Answer one question from SE . , 7,
SECTION C.) (~HTION A, one question from SECTION B, and all questions in
2. Do not write on your question paper | ee
3. Submit your
) question paper along with your answer booklet at the end of the examination.

Department Y\ dute r .

Name: ADD: tah Peace =

Matric. No. LOS 1 '56| + OD

SECTION A

Question One (Answer One Question) <

10mks a

(b) Bes the solution and interval of validity to the following IVP

eh $2t - (2 - \ln(t + 1))y' = 0$, $y(5) = 0$. 10mks '

$2y' + 5y' \geq 3y = 0$, $y(0) = -4$, $y'(0) = 9$ 12mks

(b) A 50 kg object is shot from a cannon Straight up with an initial velocity of 10m/s off a bridge that is 100 meters above the ground. If air resistance is given by $5v^*$ determine the velocity of the

mass at any time ϕ . 8mks

SECTION B | ~ ~ " 9

(Answer One Question) =

Question Three - _ "

(a) Solve the IVP $\sim |$.

$y_0 + 14y' + 49y = 0$, $y(-4) = -1$, $y'(-4) = 5$. 8mks

(b) Using Wronskian method, determine if the following functions are linearly dependent or linearly.;, $\sim \cdot \backslash - \backslash$

independent a \Y

. $f(t) = \cos t$, $g(t) = \sin t$. 6mks bs UX .

ii, $f(x) = 6^*$, $g(x) = 6^* \# 2$. 6mks a ALY eB oS

Question Four re /

(a) Solve the IVP with Laplace transforms $Y_0 > Q$,

$y'' - 6y' + 15y = 2\sin(3t)$, $y(0) = -1$, $y'(0) = -4$ 10mks Pe ety AE

(b) Solve the P.D.E. using separation of variables $\sim \sim + ')$

2 .

$\sim = k$, $u(x, 0) = f(x)$, $u(0, t) = 0$, $u(L, t) = 0$. 10mks co ® >

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Paper 3 of 3 — MTH 202 LINEAR ALGEBRA 1 2020,2021

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFKURUN,

: COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

SECOND SEMESTER EXAMINATION 2020/2021 'ef

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Course Code/Title: MU11213 Course Ville: Linear Algebra I rs) ie

Instructions; Attempt any 4 questions, Time Allowed: 2 Hours.

Question 1

x (a) Define the following subsets of Real numbers

(i) Natural numbers (ii) Whole numbers (iii) Integers (iv) Rational numbers (4 marks)

(b) Write the vector W as a linearly combination of the vectors V.,V2 and Vy. (13.5 marks)

2 3 —4 9

Vi=(1), = 2, rs=(3)) maw =(0)

3 —1 A a)

Question 2

(a) Define a vector space and list all its axioms. (6 marks)

(b) Determine if the vectors V.,V, and V3 are linearly independent. where i . (11.5 marks)

it : Zz aS)

Vy, =|2),¥.=|-2), V,=(1

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Question 3

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7 Be apeaiaes A =f Se Did

Given the following matrices A=| 3 1 /B= [i ee 5 | C= a| 6|D= [: alr

. SO ae 2 é 6 1 3)

=i 2 Ba 2

Evaluate

> t

(a) (i) AB (ii) (BA)* ey! (9 marks) Wo

(b) (i) (DC)? (ii) CD ; (8.5 marks)

Question 4

2 5 —4

(a) Find the determinant of the matrix A=|—3 6 L ity (5 marks) A

piles <2 3

(b) List 5 properties of determinants and justify each property with a solved example. (12.5 marks)

Question 5 4

(a) Find the inverse of A (A^{-1}) given that $A = \begin{bmatrix} 2 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$ (12.5 marks) a e

eal II 52. \y ;

=3j = 7

(b) Given $B = \begin{bmatrix} 5 & 2 & 1 \\ 4 & 6 & 3 \end{bmatrix}$, find the trace of the transpose of the matrix BC. (5 marks)

5 =} PD BuO

fi Question 6

\ | (a) Let $X = \mathbb{R}^2$ and $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ Define $T: X \rightarrow X$ by $Tz = Az$, where $z = \begin{pmatrix} x \\ y \end{pmatrix} \in X$. Show that T is a linear transformation and hence find the Kernel and Range of T (12.5 marks)

(b) Let $A = (2, 3, 4)$ and $B = (8, -2, 5)$ be two vectors, find the cross (vector) product of A and B ($A \times B$)

(5 marks)

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